

PCA Geometric Intuition

Dimensionality Reduction

Housing Dataset

Size of House

No. of Rooms

Price

o/p

PCA

2 dimension $\rightarrow$ 1 dimension

2D $\rightarrow$ 1D

* Loss of information
(No. of Rooms)

Feature Extraction

Eigen decomposition on matrix

Transformation

2D $\rightarrow$ 1 Dimension

much information is not lost

2 Dimension

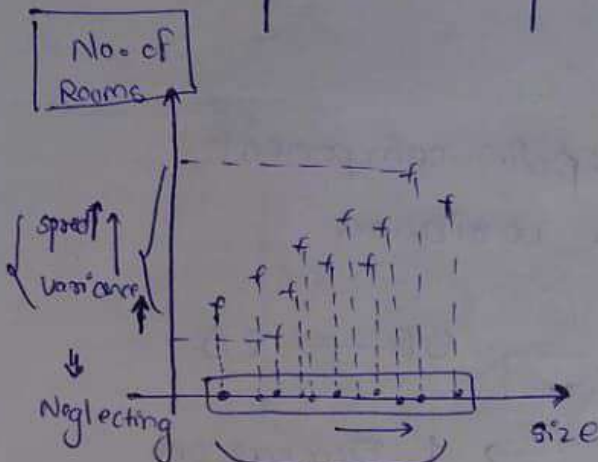
PC1, PC2

$\text{Var}(PC1) > \text{Var}(PC2)$

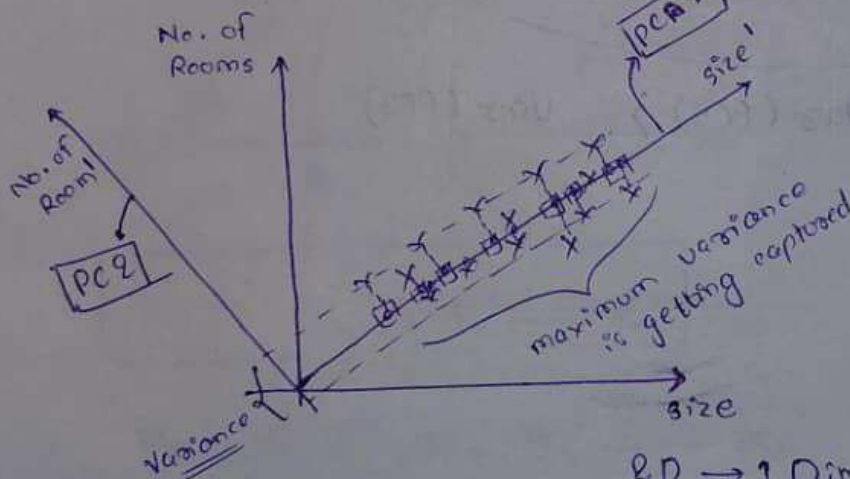
3 Dimension

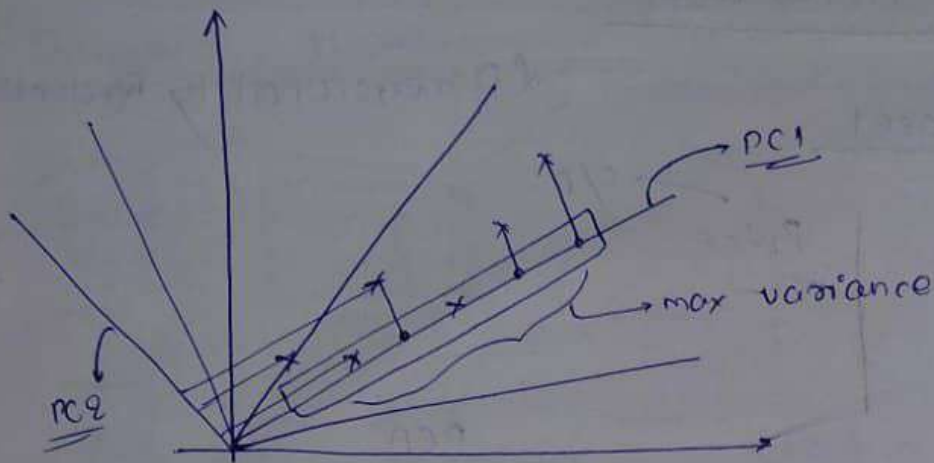
PC1, PC2, PC3

$\text{Var}(PC1) > \text{Var}(PC2) > \text{Var}(PC3)$



Spread $\uparrow$, Variance $\uparrow$



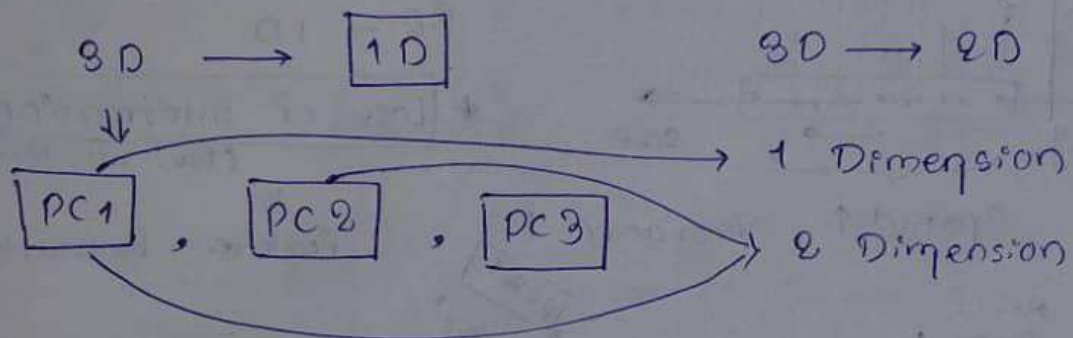


3D --- 1D

select

& Best
Principal component

To get the best Principal component
which captures maximum variance



$$\text{Var}(PC1) > \text{Var}(PC2) > \text{Var}(PC3)$$